



MDMC

Master in Data Management
and Curation



Git for beginners

Gianfranco Gallizia,
Trieste, October 1st
2026



Brief history of Git.

Version Control Systems

Git for
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Git basic
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Alias and
other tricks.

Unified diff.

Branches,
merges and
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Version Control System (VCS)

A set of software tools and procedures dedicated to keep an history of changes regarding one or more files.

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A set of software tools and procedures dedicated to keep an history of changes regarding one or more files.

For example

- SCCS - Source Code Control System (1973).

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Version Control System (VCS)

A set of software tools and procedures dedicated to keep an history of changes regarding one or more files.

For example

- SCCS - Source Code Control System (1973).
- RCS - Revision Control System (1982).

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A set of software tools and procedures dedicated to keep an history of changes regarding one or more files.

For example

- SCCS - Source Code Control System (1973).
- RCS - Revision Control System (1982).
- CVS - Concurrent Versions System (1990).

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A set of software tools and procedures dedicated to keep an history of changes regarding one or more files.

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- SCCS - Source Code Control System (1973).
- RCS - Revision Control System (1982).
- CVS - Concurrent Versions System (1990).
- SVN - Subversion (2000).

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A set of software tools and procedures dedicated to keep an history of changes regarding one or more files.

For example

- SCCS - Source Code Control System (1973).
- RCS - Revision Control System (1982).
- CVS - Concurrent Versions System (1990).
- SVN - Subversion (2000).
- Git (2005).

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VCS	Percent
Git	93,87%
SVN	5,18%
None	4,31%
Mercurial	1,13%

Fonte: Stack Overflow Developer Survey 2022

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Method of interaction	Percent
Command line	83,57%
Code Editor	54,49%
Web UI	28,44%
Dedicated GUI Application	26,37%

Fonte: Stack Overflow Developer Survey 2022

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The origins

Git was created in 2005 by Linus Torvalds following his frustration regarding the previous VCSs and the controversies regarding BitKeeper.

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Use case (1/2)

Support the development of the Linux kernel:

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Support the development of the Linux kernel:

- Huge project consisting of millions of lines of code.

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Use case (1/2)

Support the development of the Linux kernel:

- Huge project consisting of millions of lines of code.
- Thousands of developers around the World.

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Use case (1/2)

Support the development of the Linux kernel:

- Huge project consisting of millions of lines of code.
- Thousands of developers around the World.
- Everything is developed in public mailing lists.

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Support the development of the Linux kernel (cont.):

- There's no real central authority.

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Use case (2/2)

Support the development of the Linux kernel (cont.):

- There's no real central authority.
- There's no central HQ.

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Use case (2/2)

Support the development of the Linux kernel (cont.):

- There's no real central authority.
- There's no central HQ.
- There's no working time.

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Use case (2/2)

Support the development of the Linux kernel (cont.):

- There's no real central authority.
- There's no central HQ.
- There's no working time.
- There's no stable connection to a central server.

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Summarizing

- No central server.

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Summarizing

- No central server.
- The developers act independently and submit their changes to the maintainers through patches sent to the mailing list.

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Summarizing

- No central server.
- The developers act independently and submit their changes to the maintainers through patches sent to the mailing list.
- All the changes are performed locally then propagated, conflicts are solved in the integration phase.

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Summarizing

- No central server.
- The developers act independently and submit their changes to the maintainers through patches sent to the mailing list.
- All the changes are performed locally then propagated, conflicts are solved in the integration phase.
- The system has been designed by the lead maintainer of the Linux kernel and it's closer to a filesystem rather than a database of changes.

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Create an empty repository.

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Repository

Set of files and related changes plus corresponding metadata (author, date and time, optional file metadata, etc. etc.).

The inner structure of a repository is the main differentiation between VCSs.

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```
git init
```

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Create an empty repository in the current directory:

```
git init
```

Create an empty repository in the specified directory:

```
git init myrepo
```

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Create an empty repository in the current directory:

```
git init
```

Create an empty repository in the specified directory:

```
git init myrepo
```

Set the name of the main branch:

```
git init --initial-branch=trunk
```

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Set your name:

```
git config user.name "Fester Addams"
```

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Set your name:

```
git config user.name "Fester Addams"
```

Set your email:

```
git config user.email "fester.addams@example.com"
```

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Set your name:

```
git config user.name "Fester Addams"
```

Set your email:

```
git config user.email "fester.addams@example.com"
```

Setting both for each repository created by our user:

```
git config --global user.name "Fester Addams"  
git config --global user.email \  
"fester.addams@example.com"
```

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Staging area

Changes in git are not added directly to the repository, instead they are added to a buffer file (called *INDEX*) before being added to the commit tree.

The content of the INDEX together with metadata (author, timestamp, message, etc.) and the SHA-1 hash of both INDEX and metadata will become the commit.

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Staging area

Changes in git are not added directly to the repository, instead they are added to a buffer file (called *INDEX*) before being added to the commit tree.

The content of the INDEX together with metadata (author, timestamp, message, etc.) and the SHA-1 hash of both INDEX and metadata will become the commit.

From that moment on, Git will treat them as read-only data and the only way to modify them will be to destroy the commit and recreate it, altering the history of the repository and **making the modification obvious in the case of commits that have already been distributed to others.**

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Adding the files to the staging area:

```
git add README.md src/main.c python/test.py
```

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Adding the files to the staging area:

```
git add README.md src/main.c python/test.py
```

Analyze the situation:

```
git status
```

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Adding the files to the staging area:

```
git add README.md src/main.c python/test.py
```

Analyze the situation:

```
git status
```

Perform the first commit:

```
git commit -m "Initial commit."
```

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Workflow (1/2)

- 1 Make some changes.

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Workflow (1/2)

- 1 Make some changes.
- 2 Analyze the situation with:
`git status`

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Workflow (1/2)

- 1 Make some changes.
- 2 Analyze the situation with:
`git status`
- 3 Add some changes selectively:
`git add -i -p`

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Workflow (1/2)

- 1 Make some changes.
- 2 Analyze the situation with:
`git status`
- 3 Add some changes selectively:
`git add -i -p`
- 4 Review the situation:
`git status`
`git diff`
`git diff --staged`

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Workflow (2/2)

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Workflow (2/2)

- 5 Perform the partial commit (opens \$EDITOR):

```
git commit
```

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Workflow (2/2)

- 5 Perform the partial commit (opens \$EDITOR):

```
git commit
```

- 6 Add and commit all the changes (**dangerous**):

```
git add -A  
git commit -m "YOLO!"
```

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Read the commits with their relative messages:

```
git log
```

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Read the commits with their relative messages:

```
git log
```

Compact form log:

```
git log --oneline
```

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Read the commits with their relative messages:

```
git log
```

Compact form log:

```
git log --oneline
```

Single file log:

```
git log -- README.md
```

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Analyze in detail the changes made in the last commit (HEAD):

```
git show
```

```
git show HEAD
```

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Analyze in detail the changes made in the last commit (HEAD):

```
git show  
git show HEAD
```

Analyze in detail the changes made in the second-to-last commit:

```
git show HEAD~1  
git show HEAD~1
```

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Analyze a file row-by-row with commit annotations:

```
git blame -- README.md
```

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Analyze a file row-by-row with commit annotations:

```
git blame -- README.md
```

Same as before with author's email instead of author's name:

```
git blame -e -- README.md
```

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There are three default places where git searches its configuration files:

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There are three default places where git searches its configuration files:

- `$GIT_DIR/config`: valid for the current repository.

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Configuration files

There are three default places where git searches its configuration files:

- `$GIT_DIR/config`: valid for the current repository.
- `$HOME/.gitconfig`: valid for the current user.

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Configuration files

There are three default places where git searches its configuration files:

- `$GIT_DIR/config`: valid for the current repository.
- `$HOME/.gitconfig`: valid for the current user.
- `/etc/gitconfig`: valid for all the users of the local machine.

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Configuration files

There are three default places where git searches its configuration files:

- `$GIT_DIR/config`: valid for the current repository.
- `$HOME/.gitconfig`: valid for the current user.
- `/etc/gitconfig`: valid for all the users of the local machine.

Nota bene

If `/etc/gitconfig` or `$HOME/.gitconfig` do not exist git config does not return any error.

`$GIT_DIR/config` is created by `git init`.

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Configuration files format

Git configuration files are formatted like INI files.

They are divided in sections delimited by:

```
[section name]
```

options are written one per line in the following format:

```
option = value
```

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```
$HOME/.gitconfig
```

```
[user]
```

```
name = Fester Addams
```

```
email = fester.addams@example.com
```

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*.o *.pyc and other binaries

There are some files that we **DO NOT** want to place under git control:

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*.o *.pyc and other binaries

There are some files that we **DO NOT** want to place under git control:

- Compiler's output (*.o)

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*.o *.pyc and other binaries

There are some files that we **DO NOT** want to place under git control:

- Compiler's output (*.o)
- Python Bytecode (*.pyc)

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*.o *.pyc and other binaries

There are some files that we **DO NOT** want to place under git control:

- Compiler's output (*.o)
- Python Bytecode (*.pyc)
- Whole directories recreated by the build system (./bin, ./build, ...)

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`*.o *.pyc` and other binaries

We can add a file called `.gitignore` to the working directory and write in that file the patterns to ignore.

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```
.gitignore
```

```
# Ignore all .pyc
# (regardless where they are)
*.pyc

# Ignore all .o
*.o

# Ignore bin and all its content
bin/
```

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Aliases in git are like aliases in bash: sequences of characters that are expanded into predefined sequences of commands plus flags.

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Alias

Aliases in git are like aliases in bash: sequences of characters that are expanded into predefined sequences of commands plus flags.

Shorten `git checkout` in `git co`:

```
git config --global alias.co checkout
```

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Aliases in .gitconfig

It is possible to edit the `$HOME/.gitconfig` file instead of issuing the `git config` command in order to add aliases or to modify git options.

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Aliases in .gitconfig

It is possible to edit the `$HOME/.gitconfig` file instead of issuing the `git config` command in order to add aliases or to modify git options.

E.G.:

```
[user]
name = Fester Addams
email = fester.addams@example.com
[init]
defaultBranch = main
[alias]
co = checkout
ci = commit
```

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Restoring a file to the latest commit:

```
git restore -- README.md
```

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Restoring a file to the latest commit:

```
git restore -- README.md
```

Restoring all the files in the current working directory:

```
git restore .
```

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Restoring a file to the latest commit:

```
git restore -- README.md
```

Restoring all the files in the current working directory:

```
git restore .
```

Restoring all the Python source files:

```
git restore '*.py'
```

Alias e altri trucchi.

Fare pulizia

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Restoring the whole working directory:

```
git restore :/
```

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`git clean`

Sometimes you realize that your working directory is too deteriorated and you want to clean all the changes made by an automated tool or a compiler.

Other than `git restore` there's the `git clean` command.

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`git clean`

Sometimes you realize that your working directory is too deteriorated and you want to clean all the changes made by an automated tool or a compiler.

Other than `git restore` there's the `git clean` command.

CAUTION!

`git clean` deletes the files in the working directory that are **NOT** under git control.

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Dry run (does not delete):

```
git clean -n
```

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Dry run (does not delete):

```
git clean -n
```

Starting an interactive cleaning session:

```
git clean -i
```

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Dry run (does not delete):

```
git clean -n
```

Starting an interactive cleaning session:

```
git clean -i
```

Force delete (**very dangerous**):

```
git clean -f
```

```
#Also delete files present in .gitignore
```

```
git clean -f -x
```

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Comparing files

One of the most common operations performed with a VCS is comparing the differences between two revisions of one or more files.

Git uses the unified diff format, the same format you get through the `diff -u file1 file2` command and the same input format for the `patch` command.

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Header

Unified diffs start with a variable-length header. The header ends with the following two lines:

```
--- original_file  
+++ new_file
```

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Hunks

After the header there are the blocks of modified text (hunks).
A hunk has the following header:

```
@@ -line_number,length +line_number,length @@ opt
```

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Hunks

After the header there are the blocks of modified text (hunks).
A hunk has the following header:

```
@@ -line_number,length +line_number,length @@ opt
```

E.G.:

```
@@ -565,6 +565,45 @@ git clean -f -x
```

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Hunks (cont.)

After the hunk header there are the block lines. Rows starting with a - have been removed, those starting with a + have been added and those that start with a space are context lines (unchanged between the two revisions).

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Unified diff format.

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Hunks (end)

```
I'll stay here.  
-I'm going.  
+I'll arrive.
```

Unified diff.

Unified diff format.

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Putting it all together:

All I'm writing here will be ignored by patch.

Here comes the important part:

```
--- a/test.txt
+++ b/test.txt
@@ -1,2 +1,2 @@
    I'll stay here.
-I'm going.
+I'll arrive.
```

Unified diff.

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Other than `git diff` (and `git show`) git uses the unified diff format also in the following:

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Other than `git diff` (and `git show`) git uses the unified diff format also in the following:

- `git add -i -p`: Add portions of files to the staging area.

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Other than `git diff` (and `git show`) git uses the unified diff format also in the following:

- `git add -i -p`: Add portions of files to the staging area.
- `git restore -p`: Restore portions of files from the last commit.

Unified diff.

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Other than `git diff` (and `git show`) git uses the unified diff format also in the following:

- `git add -i -p`: Add portions of files to the staging area.
- `git restore -p`: Restore portions of files from the last commit.
- `git checkout -p`: Recover portions of files from a previous commit.

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Development branches

Software development is not always a linear activity, sometimes there's the need to detach temporarily or permanently from the main development line in order to progress from a common code base.

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Development branches

Software development is not always a linear activity, sometimes there's the need to detach temporarily or permanently from the main development line in order to progress from a common code base.

E.G.:

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Development branches

Software development is not always a linear activity, sometimes there's the need to detach temporarily or permanently from the main development line in order to progress from a common code base.

E.G.:

- Adding a new feature not retro-compatible.

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Development branches

Software development is not always a linear activity, sometimes there's the need to detach temporarily or permanently from the main development line in order to progress from a common code base.

E.G.:

- Adding a new feature not retro-compatible.
- Creating a new stable release that will not see new features but only bugfixes.

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Development branches

Software development is not always a linear activity, sometimes there's the need to detach temporarily or permanently from the main development line in order to progress from a common code base.

E.G.:

- Adding a new feature not retro-compatible.
- Creating a new stable release that will not see new features but only bugfixes.
- Integrating conflicting changes from other developers.

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A git commit is a data structure containing (among other things):

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A git commit is a data structure containing (among other things):

- The author's name.

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A git commit is a data structure containing (among other things):

- The author's name.
- The author's email.

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A git commit is a data structure containing (among other things):

- The author's name.
- The author's email.
- Date and time of the commit's creation.

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A git commit is a data structure containing (among other things):

- The author's name.
- The author's email.
- Date and time of the commit's creation.
- A pointer to a snapshot of the repository at that moment.

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A git commit is a data structure containing (among other things):

- The author's name.
- The author's email.
- Date and time of the commit's creation.
- A pointer to a snapshot of the repository at that moment.
- A pointer to the commit (or commits) that precede this commit.

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A git commit is a data structure containing (among other things):

- The author's name.
- The author's email.
- Date and time of the commit's creation.
- A pointer to a snapshot of the repository at that moment.
- A pointer to the commit (or commits) that precede this commit.
- The SHA-1 hash of all these metadata.

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The first commit of a repository doesn't have a preceding commit, the commits we did so far without branching and merging have saved internally the hash of the single commit preceding them.

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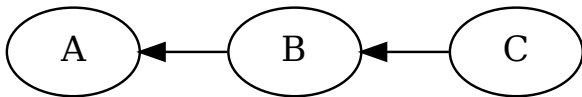
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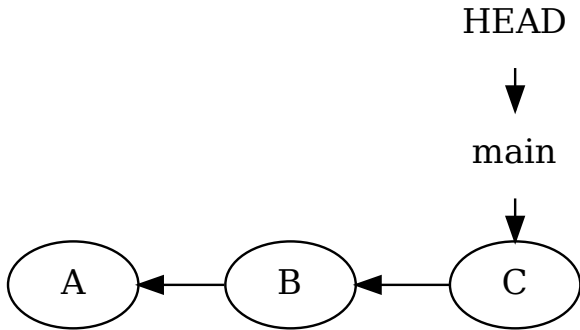
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Creating a new branch

```
git branch test
```

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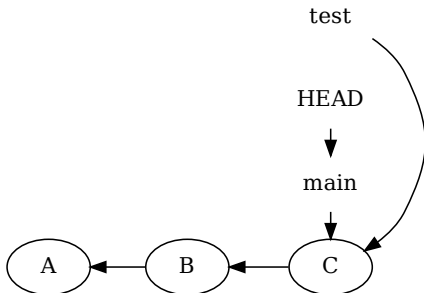
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Creating a new branch

```
git branch test
```



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Moving to the new branch

```
git checkout test
```

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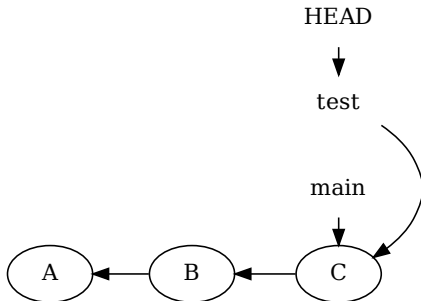
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```
git checkout test
```



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Making some changes and committing them:

```
git add -- fancy.py
git commit -m "Added fancy Python script."
```

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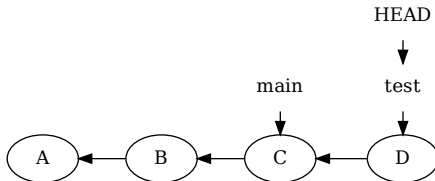
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Making some changes and committing them:

```
git add -- fancy.py
```

```
git commit -m "Added fancy Python script."
```



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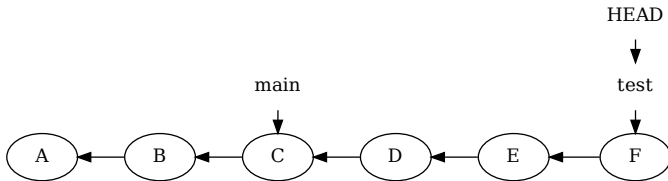
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After other commits. . .



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Going back to main:

```
git checkout main
```

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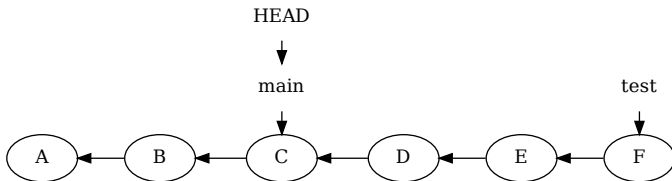
Alias and
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Unified diff.

Branches,
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conflicts.

Going back to main:

```
git checkout main
```



Branches, merges and conflicts.

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Integrating the changes:

```
git merge test
```

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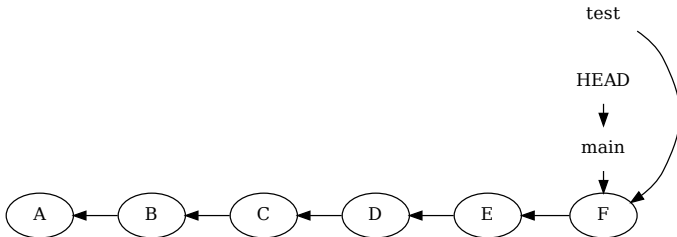
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Integrating the changes:

```
git merge test
```



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Let's get back before the merge. . .

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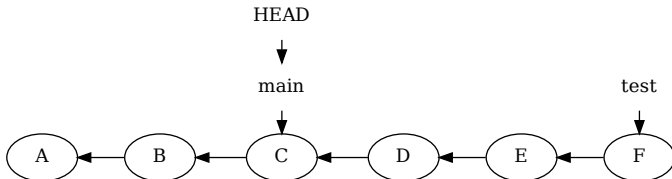
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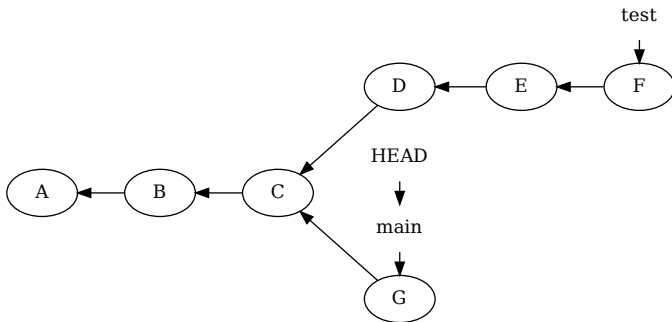
Branches,
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Let's commit a hotfix on `main`...

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Let's commit a hotfix on `main`...



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At this point a *fast-forward* is no longer possible, we need to create a merge commit: a commit that has both F and G as parents and that collects the changes from both branches.

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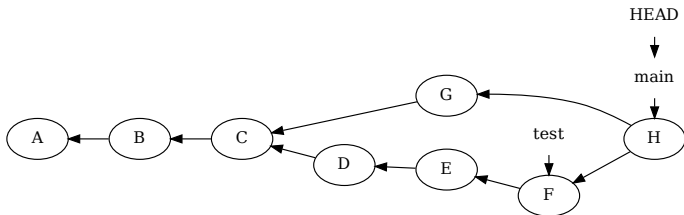
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At this point a *fast-forward* is no longer possible, we need to create a merge commit: a commit that has both F and G as parents and that collects the changes from both branches.



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Conflicts

Let's assume we're in a situation where a conflict arises. How can we manage it when merging two branches in git?

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Conflicts

Let's assume we're in a situation where a conflict arises. How can we manage it when merging two branches in git?

```
git init --initial-branch=main conflict
cd conflict
echo -e "A\nB\nC" > README.txt
git add README.txt
git commit -m "Initial Commit."
```

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Conflicts (cont.)

```
git checkout -b change_A
tr A-Z a-z < README.txt > tmp.txt
mv tmp.txt README.txt
git add README.txt
git commit -m "Lowercase."
```

Branches, merges and conflicts.

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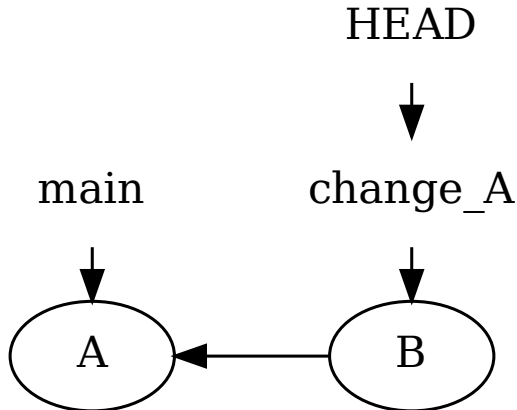
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Conflicts (cont.)

```
git checkout main
sed -i -r 's/^(([A-Z]))/\1\1/' README.txt
git add README.txt
git commit -m "Double the letters."
```

Branches, merges and conflicts.

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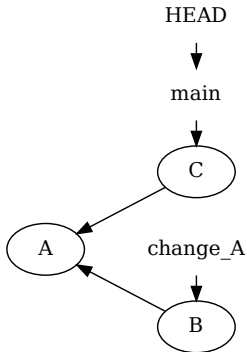
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Conflicts (cont.)

```
git checkout -b change_B
```

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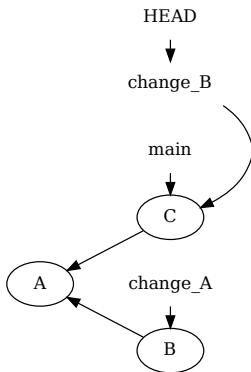
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Unified diff.

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conflicts.

Conflicts (cont.)

```
git checkout -b change_B
```



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Conflicts (cont.)

```
git merge change_A
```

```
Auto-merging README.txt
```

```
CONFLICT (content): Merge conflict in README.txt
```

```
Automatic merge failed; fix conflicts and then commit
```

Branches, merges and conflicts.

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First let's check the status:

```
git status
```

```
On branch change_B
```

```
You have unmerged paths.
```

```
    (fix conflicts and run "git commit")
```

```
    (use "git merge --abort" to abort the merge)
```

```
Unmerged paths:
```

```
    (use "git add <file>..." to mark resolution)
```

```
both modified:   README.txt
```

```
no changes added to commit
```

```
(use "git add" and/or "git commit -a")
```

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Let's check the changes in detail:

```
git diff -- README.txt
diff --cc README.txt
index 21c9ecf,de98044..0000000
--- a/README.txt
+++ b/README.txt
@@@ -1,3 -1,3 +1,9 @@@
++<<<<<< HEAD
+AA
+BB
+CC
++=====
+ a
+ b
+ c
++>>>>>> change_A
```

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README.txt

<<<<<<< HEAD

AA

BB

CC

=====

a

b

c

>>>>>>> change_A

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README.txt

aA

bB

cC

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Let's resolve the conflict

```
git commit -a
git log --all --decorate --oneline --graph
* 8b393c8 (HEAD -> change_B) Merge branch...
|\
| * 5855fe7 (change_A) Lowercase.
* | bd1130e (main) Double the letters.
|/
* d1f0f78 Initial Commit.
```

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Let's resolve the conflict

```
git diff main
diff --git a/README.txt b/README.txt
index 21c9ecf..a359da0 100644
--- a/README.txt
+++ b/README.txt
@@ -1,3 +1,3 @@
-AA
-BB
-CC
+aA
+bB
+cC
```

Branches, merges and conflicts.

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Let's resolve the conflict

```
git diff change_A
diff --git a/README.txt b/README.txt
index de98044..a359da0 100644
--- a/README.txt
+++ b/README.txt
@@ -1,3 +1,3 @@
-a
-b
-c
+aA
+bB
+cC
```

End of module 1

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Thank you for your attention!

End of module 1

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Thank you for your attention!

Any questions?



OGS

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e di Geofisica
Sperimentale

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